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# Standard Operating Procedure

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VERSION

0.1 DRAFT

STATUS

DRAFT

OWNER

Hawkeye Platform

RENDERED

2026-06-16

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SOP-MFG-001 · Active Pharmaceutical Ingredient Manufacturing

| FIELD          | VALUE                                           |
|----------------|-------------------------------------------------|
| Document Owner | Quality Assurance · Sai Life Sciences Pvt. Ltd. |
| SOP Number     | SOP-MFG-001                                     |
| Version        | 04                                              |
| Effective Date | 2026-04-01                                      |
| Next Review    | 2027-04-01                                      |
| Approval       | QA Head · Production Head · Site Head           |
| Applies to     | API manufacturing block 2 · Hyderabad facility  |

1. Purpose

This Standard Operating Procedure (SOP) defines the controlled process for manufacturing Active Pharmaceutical Ingredients (APIs) at Sai Life Sciences' Hyderabad facility in accordance with **ICH Q7** Good Manufacturing Practice for Active Pharmaceutical Ingredients, **21 CFR 211** Current Good Manufacturing Practice for Finished Pharmaceuticals, and **EU GMP Part II** (chapters 1-19).

2. Scope

Applies to manufacturing of non-sterile APIs in production block 2 covering reaction, isolation, drying, and bulk packaging unit operations. Excludes sterile API manufacturing (governed by SOP-MFG-002).

### 3. Responsibilities

| ROLE                | RESPONSIBILITY                                                    |
|---------------------|-------------------------------------------------------------------|
| Production Operator | Execute batch instructions; record observations contemporaneously |
| Shift Supervisor    | First-line verification of in-process checks                      |
| QA In-Process       | Sampling oversight; phase-gate verification                       |
| QA Batch Release    | Final disposition; CoA review                                     |
| QC                  | Analytical testing per validated methods                          |

### 4. Procedure (high-level)

1. **Batch initiation:** Operator retrieves controlled Master Batch Record (MBR) version 4.2; QA verifies revision currency against the controlled-doc index.
2. **Material verification:** Each charged raw material's CoA is verified against approved supplier list. Material identity confirmed by IR or HPLC per material spec.
3. **Process execution:** Each unit operation completed per validated process parameters logged in BMR; deviations captured in deviation system.
4. **In-process controls:** IPC samples drawn at each phase transition per sampling plan; results entered in LIMS within 4 hours.
5. **Phase gating:** No phase progression until prior phase IPC results meet acceptance criteria; deviations require QA-approved investigation.
6. **Drying & milling:** Per equipment-specific SOPs; particle-size analysis performed pre and post milling.
7. **Bulk packaging:** Tared drums; net weight verified against batch theoretical yield; label per labelling SOP-LBL-007.
8. **Batch closure:** BMR review by Production then QA; closure within 30 calendar days of last operation.
9. **Release decision:** QA Batch Release reviews complete BMR, CoA, deviations, change controls; documented release decision in QMS.

### 5. Records & retention

| RECORD                     | RETENTION                      |
|----------------------------|--------------------------------|
| Batch Manufacturing Record | 1 year past expiry + 5 years   |
| In-process control records | 1 year past expiry + 5 years   |
| Equipment logs             | 5 years from last entry        |
| Deviation reports          | Permanent                      |
| Training records           | Personnel-employment + 5 years |

### 6. References

- ICH Q7 · Section 6 (Documentation and Records), Section 8 (Process Controls)
- 21 CFR 211.100–211.115
- EU GMP Part II, Chapters 5 (Production) and 6 (Quality Control)
- Sai Life Sciences Quality Manual QM-001 v3
- Sai Life Sciences Validation Master Plan VMP-2024

## 7. Revision history

| VERSION | DATE       | CHANGE                                                                       |
|---------|------------|------------------------------------------------------------------------------|
| 01      | 2022-06-15 | Initial issue                                                                |
| 02      | 2023-08-10 | Added IR identity verification (regulatory observation closure)              |
| 03      | 2024-11-22 | Extended retention for deviation reports to "Permanent" per MHRA observation |
| 04      | 2026-04-01 | LIMS integration for IPC results (was paper-form) · 4-hr entry SLA added     |

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